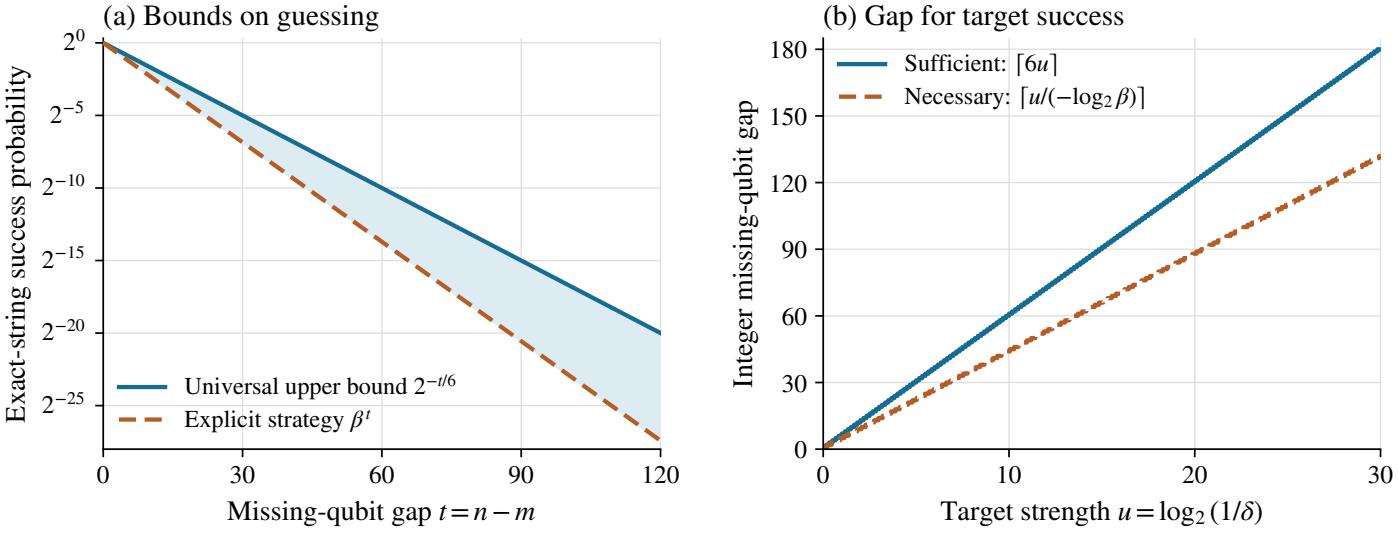
(a) Bounds on guessing
Exact-string success probability
$2^{0}$
$2^{-5}$
$2^{-10}$
$2^{-15}$
$2^{-20}$
$2^{-25}$
Universal upper bound $2^{-t/6}$
Explicit strategy $\beta^{t}$
0
30
60
90
120
Missing-qubit gap $t = n - m$
(b) Gap for target success
Integer missing-qubit gap
Sufficient: $\lceil 6u \rceil$
Necessary: $\lceil u/(-\log_2 \beta) \rceil$
180
150
120
90
60
30
0
0
10
20
30
Target strength $u = \log_2(1/\delta)$